

4 (a) $\Delta U = q + w$ (allow correct word equation) B1 [1]

(b) *either* kinetic energy constant because temperature constant M1
potential energy constant because no intermolecular forces M1
so no change in internal energy A1 [3]
or kinetic energy and potential energy both constant (M1)
so no change in internal energy (A1)
reason for *either* constant k.e. or constant p.e. given (A1)

5 (a) change/loss in kinetic energy = change/gain in electric potential energy B1